

Evaluation of Low-Cost sensors using Distributed Monitoring System on a CNC Machine

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Abstract—This work presents the evaluation of a low-cost distributed monitoring system for CNC machine tools. The proposed platform combines Arduino-based acquisition units, a Raspberry Pi, electrical current sensors, accelerometers, vibration sensors, and acoustic sensors to observe the electrical and mechanical behavior of the machine under different operating and machining conditions. Communication, acquisition, and data-management tasks are coordinated through a Multi-Agent Distributed System (MADS), while the measurements and experimental metadata are stored in MongoDB.

The system was deployed during three experimental campaigns involving variations in spindle speed, feed rate, cutting depth, radial cutting depth, and cutting direction. Electrical currents, three-axis accelerations, RMS vibration, and acoustic signals were recorded simultaneously to provide a multi-sensor representation of machine operation.

Index Terms—CNC machine, condition monitoring, MongoDB, Python, Arduino, Raspberry Pi, MADS, vibration, current sensing, acoustic sensing.

I. OBJECTIVE

The objective of this work was to evaluate a low-cost monitoring solution for CNC milling machines based on the Multi-Agent Distributed System (MADS) developed by Professor Paolo Bossetti at the University of Trento [1].

To determine whether the monitoring system can identify meaningful output variations in CNC milling, the experimental campaign compared the machine's electrical, mechanical, and acoustic responses under different operating conditions.

The study also assessed the system's potential for industrial implementation. This requires reliable, repeatable, and interpretable measurements while preserving low cost, distributed operation, flexibility, and ease of installation.

II. METHODOLOGY

A. Experimental set-ups

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B. Materials and Equipment

The experimental platform consisted of a CNC milling machine, cutting tools and inserts, low-cost sensors, Arduino acquisition boards, a Raspberry Pi, and the computing resources required to operate the distributed monitoring architecture.

Figure 1 shows the Mikron 500 CNC milling machine used during the experimental campaigns and a close-up of the milling cutter equipped with the corresponding cutting inserts.



Fig. 1. (A) Mikron 500 CNC milling machine used during the experimental campaigns; (B) close-up of the indexable milling cutter equipped with Sandvik Coromant R390-11 T3 08M-PM 4340 inserts.

C. Accelerometer Mounting Support

A custom mechanical support, shown in figure 2, was designed by Antonio Festas, workshop technician at the University of Aveiro, to mount the accelerometer on the machine spindle assembly. The support was developed to maintain the sensor in a stable and repeatable position during the milling tests. A rigid attachment was required to reduce relative movement between the accelerometer and the machine structure and to ensure that the measured acceleration was representative of the spindle response. The support also maintained a consistent sensor orientation throughout the experimental campaigns, allowing the acceleration signals obtained under different machining conditions to be compared.

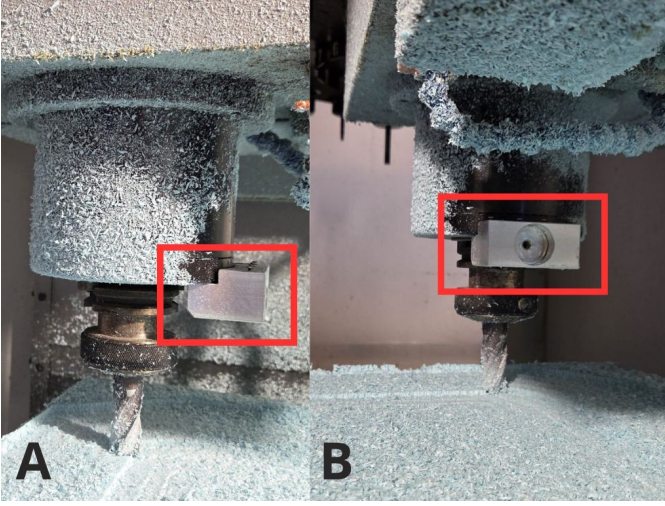


Fig. 2. Custom accelerometer mounting support installed on the CNC spindle assembly: (A) lateral view and (B) frontal view. The support is highlighted by the red rectangles.

D. Architecture

The monitoring platform combines a physical acquisition layer with a distributed communication and data-management layer based on MADS. The physical architecture describes the connections between the sensors, Arduino acquisition boards, Raspberry Pi, and monitoring computer. The MADS architecture describes how the acquired measurements are exchanged, visualized, and stored.

1) *Physical Acquisition Architecture*: The physical acquisition architecture is presented in Fig. 3. The sensing system was divided between two Arduino UNO boards, each responsible for acquiring a different group of signals.

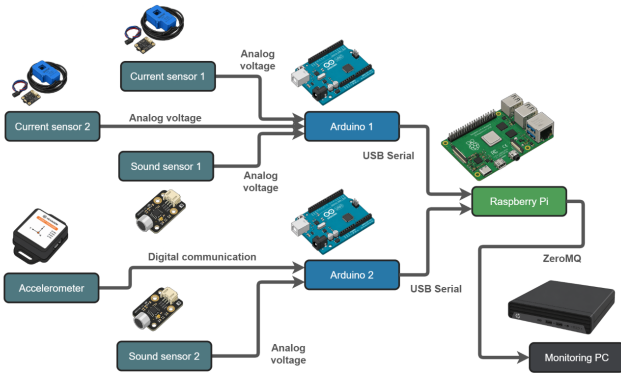


Fig. 3. Physical architecture of the CNC monitoring platform. The sensors are connected to two Arduino UNO acquisition boards, which transmit the measurements to the Raspberry Pi through USB serial communication. The Raspberry Pi communicates with the monitoring computer through the network connection.

Table I summarizes the main components used during the experimental campaigns and their respective functions.

TABLE I
MATERIALS AND EQUIPMENT USED IN THE EXPERIMENTAL PLATFORM

Component	Reference	Function
CNC milling machine	Mikron 500	Execution of the milling experiments
Milling cutter	Sandvik Coromant CoroMill 390	Material removal during the milling operations
Cutting inserts	Sandvik Coromant R390-11 T3 08M-PM 4340	Replaceable cutting elements used during machining
Acquisition boards	Arduino UNO R4	Acquisition and transmission of sensor measurements
Central computing unit	Raspberry Pi 5 model B	Coordination of the distributed monitoring architecture
Current sensors	DFRobot SEN0211	Measurement of the machine and motor electrical currents
Vibration sensor	SW-420	Detection of vibration during machine operation
Acoustic sensors	DFRobot DFR0034	Acquisition of acoustic signals during milling
Monitoring computer	HP Pro Mini 400 G9 Desktop PC	Execution of monitoring, visualization, database access, and data-analysis tasks

Arduino 1 acquired the following measurements:

- machine current from a SEN0211 current sensor;
- motor current from a second SEN0211 current sensor;
- the first acoustic signal from a DFR0034 sound sensor.

The SEN0211 current sensors and the DFR0034 sound sensor generated analog voltage signals, which were connected to the analog inputs of Arduino 1.

Arduino 2 acquired the following measurements:

- three-axis acceleration from the accelerometer;
- the second acoustic signal from a DFR0034 sound sensor.

The accelerometer communicated digitally with Arduino 2, whereas the second DFR0034 sensor generated an analog voltage signal.

Both Arduino boards transmitted the acquired measurements to the Raspberry Pi through USB serial communication. The Raspberry Pi acted as the central computing node of the experimental platform and established the communication link with the monitoring computer.

2) *MADS Communication and Data Architecture*: The data architecture is presented in Figure 4. On the Raspberry Pi, the Arduino plugin receives the measurements transmitted by the acquisition boards and transfers them to a MADS source agent.

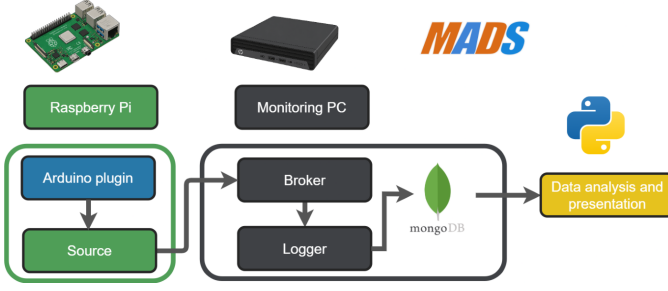


Fig. 4. MADS communication and data architecture. The Raspberry Pi runs the Arduino plugin and the source agent, which publishes the acquired measurements to the broker. The logger records the messages in MongoDB, and Python is used for data analysis and presentation.

The source agent publishes the measurements through the MADS communication system. The broker routes the messages to the logger, which stores the sensor data, timestamps, and associated metadata in MongoDB. Communication between the distributed MADS components is implemented through ZeroMQ over TCP/IP.

The monitoring PC hosts the main data-management services and provides access to the database. The stored measurements are subsequently retrieved with Python for test reconstruction, signal analysis, and graphical presentation.

3) *Experimental Assembly*: The complete experimental assembly included the sensors installed on the CNC machine, the two Arduino acquisition boards, the Raspberry Pi, the monitoring computer, and the corresponding power, signal, serial, and network connections.

The current sensors were installed around the conductors associated with the machine and motor electrical supplies. The accelerometer was attached to the spindle assembly using the custom mechanical support described previously, while the acoustic sensors were positioned close to the machining area.

The Arduino boards and Raspberry Pi were placed outside the working area to protect the electronic components from chips, cutting fluid, and mechanical contact. The final arrangement also allowed the connections to remain accessible during configuration and testing.

III. RESULTS

A. Experimental Campaign Overview

The three experimental sessions generated a total of 76 independent test intervals. Table II summarizes the number and type of tests obtained during each session, together with the ranges of the principal operating and machining parameters.

TABLE II
SUMMARY OF THE EXPERIMENTAL CAMPAIGNS

Date	Tests	Test type	Parameter ranges
Sept. 11	16	Milling tests	Spindle speed: 2000–5000 rpm; feed rate: 500–3000 mm/min; axial cut depth: 1.0 mm; radial cut depth: 70%; cut length: 90 mm; direction: upmilling and downmilling; duration: 12.7–30.5 s.
Sept. 16	32	Segmented spindle-speed intervals	Spindle speed: 50–7500 rpm;
Sept. 16	12	Milling tests	Spindle speed: 2000–3500 rpm; feed rate: 500–1000 mm/min; axial cut depth: 0–1 mm; radial cut depth: 2 mm; cut length: 75 mm; direction: upmilling and downmilling.
Sept. 17	16	Milling tests	Spindle speed: 2500 rpm; feed rate: 500–2000 mm/min; axial cut depth: 2.0 mm; radial cut depth: 2.0 mm; cut length: 75 mm; direction: upmilling and downmilling; duration: 10.0–25.4 s.

The detailed parameter combination corresponding to each individual test is presented in its graphical summary and in the complete results of the respective experimental session.

B. Session of September 11, 2025

The measurements obtained on September 11 were organized as 16 independent tests. For each test, the available electrical, vibration, acceleration, and acoustic measurements were plotted against relative time.

Figure 5 presents a representative graphical summary. The standardized format facilitates the comparison of the measured responses under different machining conditions.

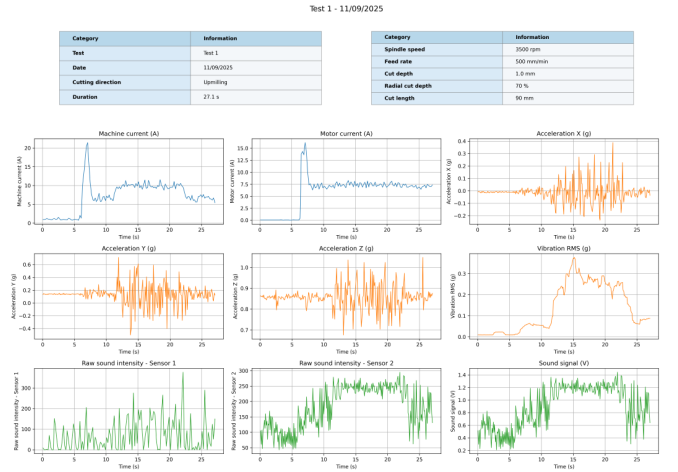


Fig. 5. Representative graphical summary of a test conducted on September 11, 2025.

C. Experimental Session of September 16, 2025

Unlike the sessions of September 11 and 17, part of the data recorded on September 16 contained complex intervals in which several operating conditions were grouped within a single metadata-defined test. In particular, some intervals included successive spindle-speed levels within the same acquisition window. For this reason, the results of this date are presented through a representative example showing the original complex test, the cutting procedure used to separate it, and the final result obtained for one segmented test.

1) *Example of a Complex Test:* Figure 6 presents an example of a complex test recorded on September 16, 2025. In this interval, the monitored signals correspond to several successive operating conditions rather than to a single stable test. This type of recording required additional processing before the data could be analyzed as independent experiments.

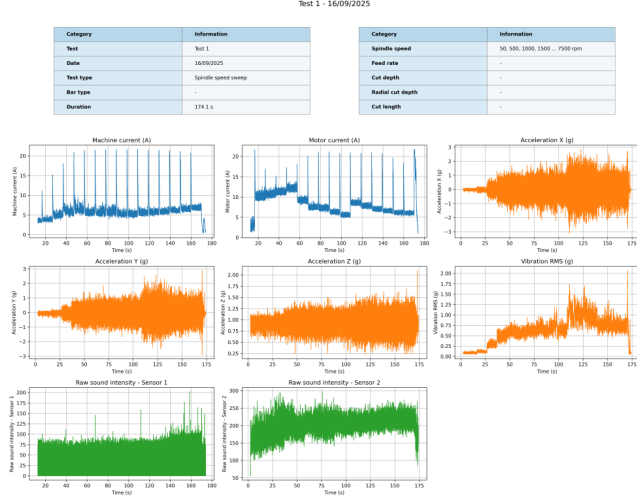


Fig. 6. Example of a complex test recorded on September 16, 2025, containing several operating stages within the same acquisition interval.

2) *Cutting Procedure:* To transform the complex interval into independent and usable tests, a segmentation procedure was applied. The transitions between operating stages were identified from the measured signals, allowing the original interval to be divided into separate sub-tests corresponding to different spindle-speed levels or machining conditions.

Figure 7 illustrates the cutting procedure used for this purpose.

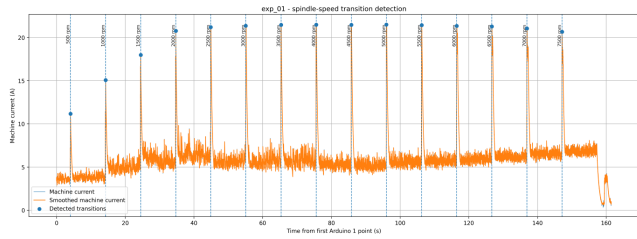


Fig. 7. Illustration of the segmentation procedure applied to a complex test recorded on September 16, 2025. The detected transitions were used to divide the original interval into independent tests.

3) *Example of a Segmented Test Result:* After segmentation, each resulting interval was treated as an independent test and processed using the same visualization procedure as the other experimental sessions. Figure 8 presents the graphical summary of one representative segmented test obtained from the original complex interval.

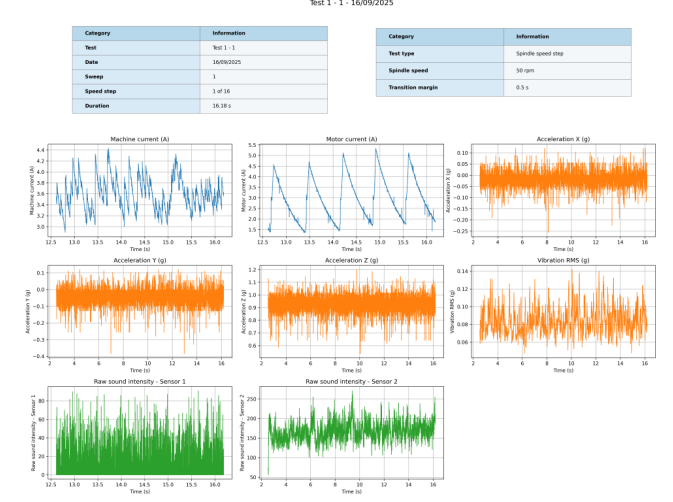


Fig. 8. Example of an individual test obtained after segmenting a complex interval recorded on September 16, 2025.

This procedure made it possible to convert complex acquisition intervals into independent and interpretable tests, which could then be compared with the remaining experiments of the campaign.

D. Experimental Session of September 17, 2025

The tests conducted on September 17 were reconstructed from the metadata markers and associated with their machining parameters.

The same procedure was used to retrieve the sensor measurements, calculate relative time, identify the available signals, and generate standardized graphical summaries.

Figure 9 presents the respective representative graphical summary.

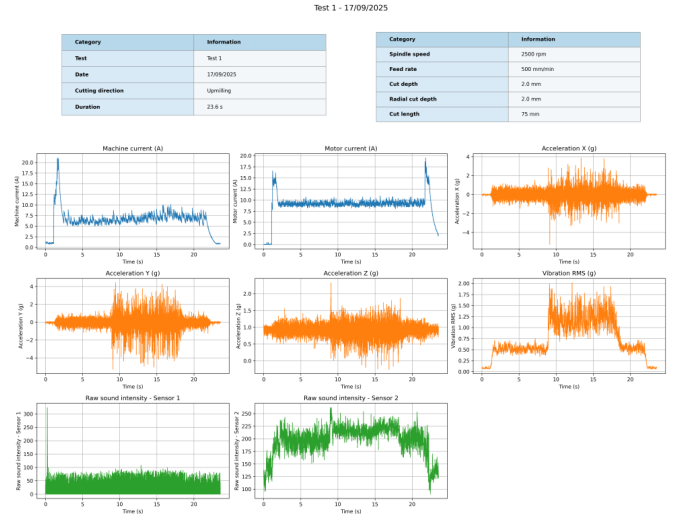


Fig. 9. Representative graphical summary for a test conducted on September 17, 2025.

E. General Signal Behavior

The standardized summaries reveal several operating phases in the acquired signals:

- machine start-up;
- spindle-speed transitions;
- entry into machining;
- approximately stable operating periods;
- changes in vibration and acoustic activity;
- machine deceleration and stopping.

The machine-current and motor-current signals are particularly useful for identifying operating transitions.

The acceleration and RMS-vibration measurements provide information about the mechanical response of the machine.

The acoustic signals provide an additional observation channel that may complement the electrical and vibration measurements.

IV. CONCLUSION

ACKNOWLEDGMENT

The author acknowledges INSA Toulouse, the Institut Clément Ader (ICA), the Maison de la Formation Jacqueline Auriol (MFJA), the University of Aveiro, the University of Trento, and the European Consortium of Innovative Universities (ECIU) for providing the facilities, equipment, technical resources, and institutional support used in this project.

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